

BASICS FOR AIR TRAFFIC CONTROL – AIRSPACE

INTRODUCTION

When you take a trip in your car, there are many regulations that dictate safe travel. These regulations are posted on the road's pavement or as signs along your route.

However, you cannot post a sign in a cloud or mark the lanes of travel in the air. The National Airspace System (NAS) must be highly organized and regulated to operate safely. Imagine thousands of highways extended in the sky, layered vertically and horizontally, and intersecting in a complicated web. Picture having to avoid traffic on these highways in three-dimensional airspace. Traffic is not only to your right and left, but above and below you, too.



Air Traffic Control (ATC) manages such a complex network by classifying airspace and defining its dimensions.

The purpose of this module is to explain how Air Traffic Control (ATC) classifies, organizes, and uses classes of airspace and Special Use Airspace (SUA).

CLASSES OF AIRSPACE

Purpose: This lesson explains how airspace is classified and organized by Air Traffic Control (ATC) and how ATC uses the airspace to safely direct and separate thousands of aircraft flights each day.

Objective:

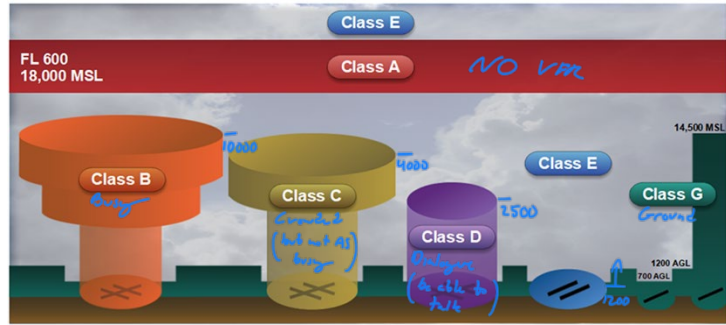
- Identify classes of airspace and their uses

References for this lesson are as follows:

- Title 14 Code of Federal Regulations, Part 91
- FAA Order JO 7110.65, Air Traffic Control
- FAA Order JO 7400.2, Procedures for Handling Airspace Matters
- Aeronautical Information Manual (AIM)

Classes of Airspace

Controlled airspace has defined dimensions. ATC provides Air Traffic Services (ATS) within these dimensions for Instrument Flight Rules (IFR) flights and Visual Flight Rules (VFR) flights depending on the airspace classification. Airspace classes include Classes A, B, C, D, and E airspace where all aircraft operators are subject to certain pilot qualifications, operating rules, and equipment requirements.



Controlled

	Class A	Class A airspace is generally that airspace from 18,000 feet Mean Sea Level (MSL) up to and including FL 600 (Flight Level Six Zero Zero).
	- All operations must be conducted under IFR and are subject to ATC clearances and instructions - Two-way radio communication must be established and maintained - A 4096 transponder with functioning Mode C is required - ADS-B transponder - Pilot must be instrument rated and aircraft must be IFR equipped and certified - Jet routes are contained in Class A airspace	
Everyone gets separated	Class B	Class B is generally that airspace from the surface to 10,000 feet MSL surrounding the nation's busiest airports in terms of airport operations or passenger enplanements.
	Configuration: - Individually tailored and consists of a surface area and two or more layers - Designed to contain all published instrument approach procedures once an aircraft enters the airspace To operate in Class B airspace: - All aircraft require an ATC clearance to enter - ATC provides separation services; therefore, two-way radio communication must be established and maintained - A 4096 transponder with functioning Mode C is required - ADS-B transponder - IFR aircraft require operable VOR or TACAN navigation equipment All aircraft (VFR or IFR) receive separation services while in this airspace. Student pilots may NOT operate an aircraft on a solo flight unless: - Student pilot has received instruction on that specific Class B airspace - Student pilot's logbook has been endorsed within the preceding 90 days verifying instruction	



Class C

Class C is generally that airspace from the surface to 4,000 feet above the airport elevation (charted in Mean Sea Level [MSL]) surrounding those airports that have an operational control tower, service provided by a Radar Approach Control, and a certain number of IFR operations or passenger enplanements.

Configuration:

- Individually tailored and usually consists of a surface area with a 5 NM radius
- A circle with a 10-NM radius that extends no lower than 1,200 feet up to 4,000 feet above the airport elevation
- An outer area that is not charted
 - Non-regulatory airspace surrounding designated Class C airspace airports wherein ATC provides radar vectoring and sequencing on a full-time basis for all IFR and participating VFR aircraft

To operate in Class C Airspace:

- ATC provides separation services; therefore, two-way radio communication must be established and maintained
 - VFR aircraft do not require a clearance to enter, but entry may be denied
- Two-way communications, ADS-B transponder and a 4096 transponder with functioning Mode C are required
- VFR aircraft are separated only from IFR aircraft, and IFR aircraft are separated from all other aircraft

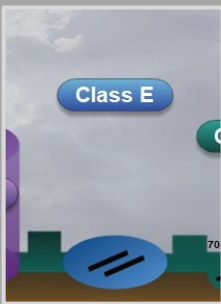


Class D

Class D is generally that airspace from the surface to 2,500 feet above the airport elevation (charted in MSL) surrounding those airports that have an operational control tower.

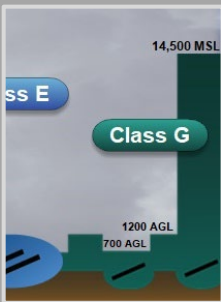
Configuration:

- Individually tailored
- Normally designed to contain published instrument approaches
- ATC provides separation between IFR aircraft
- No separation services are provided to VFR aircraft.

	Class E Generally, if airspace is NOT Class A, Class B, Class C, or Class D, and it is controlled airspace, it is Class E. Except for 18,000 feet MSL, Class E airspace has NO defined vertical limit, but rather, it extends upward from either the surface or a designated altitude to the overlying or adjacent controlled airspace. Federal airways (“Victor”) and low altitude RNAV routes (“Tango”) are Class E airspace and, unless otherwise specified, extend upward from 1,200 feet Above Ground Level (AGL) to, but NOT including, 18,000 feet MSL. When designated as a surface area, the airspace will be configured to contain all instrument procedures. There is NO communication requirement or separation provided for VFR aircraft. Generic controlled airspace consisting of airways and controlled areas between airways. To qualify for Class E surface area, the airport must have weather observation and reporting capabilities.
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Note: There are additional requirements for ADS-B equipment in Class E airspace over the Gulf of Mexico from the coastline of the United States out to 12 NM.

Uncontrolled Airspace

	Class G Class G airspace (uncontrolled) is that portion of airspace that has NOT been designated as Class A, Class B, Class C, Class D, or Class E airspace. Class G airspace normally extends from the surface upwards to the base of the overlying controlled airspace. Vector aircraft in Class G airspace only upon pilot request and as an additional service.
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Note: While separation standards do not exist in certain airspace classes for VFR aircraft, the primary purpose of the ATC system is to prevent collisions between aircraft operating in the system. Controllers shall give first priority to separating aircraft and issuing safety alerts.



Knowledge Check A

REVIEW what you have learned so far about the classes of airspace. ANSWER the questions listed below.

1. Which class of airspace does **NOT** require an ATC clearance? (Select the correct answer.)

- ☐ Class A
- ☐ Class B
- ☒ Class C

2. The airspace that generally extends from the surface to 10,000 feet MSL and surrounds a busy airport is designated as _____ airspace. *(Select the correct answer.)*
 - ☐ Class A
 - ☒ Class B
 - ☐ Class C
3. Which class of airspace is uncontrolled? *(Select the correct answer.)*
 - ☐ Class D
 - ☐ Class E
 - ☒ Class G
4. Which class of airspace is generally that airspace from the surface to 2,500 feet above the airport elevation? *(Select the correct answer.)*
 - ☐ Class E
 - ☒ Class D
 - ☐ Class G
5. Which airspace has **NO** defined vertical limit? *(Select the correct answer.)*
 - ☐ Class D
 - ☒ Class E
 - ☐ Class B

Temporary Flight Restrictions

Temporary Flight Restrictions (TFRs) are published via Flight Data Center Notice to Airmen (FDC NOTAM). TFRs are issued within the sovereign airspace of the United States and its territories to restrict certain aircraft from operating within a defined area on a temporary basis to protect persons or property in the air or on the ground.

TFRs may be issued for:		
		
Aircraft relief activities following a disaster	Aircraft accident/incident sites	Disaster or hazardous situations
		
Aviation or ground resources engaged in wildfire suppression	Support of VIP movements	Reasons of national security



Knowledge Check B

REVIEW what you have learned so far about TFRs. ANSWER the question listed below.

1. Why are TFRs issued? *(Select the correct answer.)*
 - ☐ To restrict all aircraft from operating within a defined area on a permanent basis
 - ☒ To restrict certain aircraft from operating within a defined area on a temporary basis
 - ☐ To restrict certain aircraft from operating within a defined area on a permanent basis

Classes of Airspace Summary

This lesson explained how airspace is classified, organized, and operated by ATC and the purposes of TFRs.

STUDY AID: AIRSPACES QUICK REFERENCE CHART

AIRSPACE FEATURES	CLASS					
	A	B	C	D	E	G
Entry Requirements	IFR Clearance	ATC Clearance	Prior two-way communications	Prior two-way communications	None	None
Minimum Pilot Qualifications	Instrument rating	Private or student certificate location dependent	Student certificate	Student certificate	Student certificate	Student certificate
Two-Way Radio Communications	Yes	Yes	Yes	Yes	Not required	Not required
Special VFR Allowed*	No	Yes	Yes	Yes	Yes	N/A
VFR Aircraft Separation	N/A	All	IFR	Runway operations	None	None
Traffic Advisories	N/A	Yes	Yes	Workload permitting	Workload permitting	Workload permitting

* Authorized by an ATC clearance and conducted within the lateral boundaries of the surface area.

Note: IFR operations in controlled airspace require filing an IFR flight plan and an appropriate ATC clearance.

SPECIAL USE AIRSPACE

Purpose: This lesson defines Special Use Airspace (SUA), enabling air traffic controllers to enforce regulations concerning the nature of activities held within these reserved areas.

Objective:

- Identify SUA, as designated in the United States

References for this lesson are as follows:

- FAA Order JO 7110.65, Air Traffic Control
- FAA Order JO 7400.2, Procedures For Handling Airspace Matters

Special Use Airspace

Special Use Airspace (SUA) is airspace of defined dimensions identified by an area on the surface of the Earth wherein activities must be confined because of their nature and/or wherein limitations may be imposed upon aircraft operations that are not a part of those activities.



Special Use Airspace Programs

SUA programs are designed to:

- **Accommodate** National Defense, security, and welfare and necessary military activity
- **Identify** where activity occurs
- **Protect** other users from hazardous operations

Controlling Agency and Using Agency

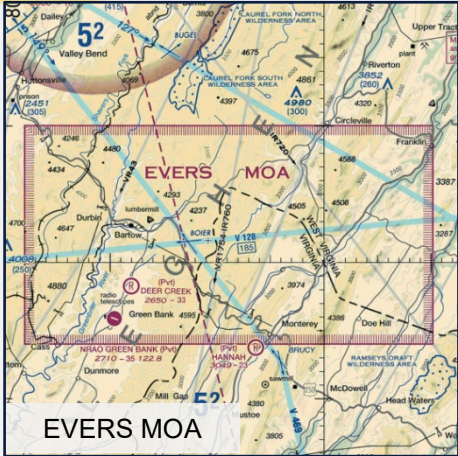
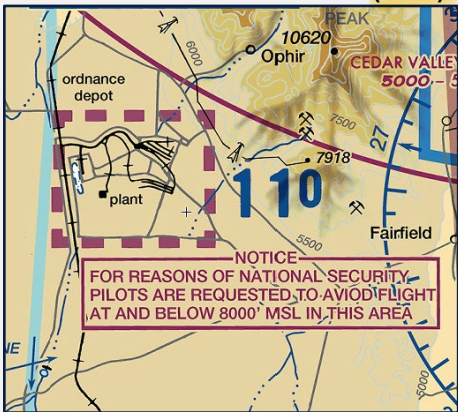
CONTROLLING AGENCY	USING AGENCY
The controlling agency is the FAA ATC facility that <u>exercises control of the airspace</u> when an SUA area is NOT activated. ■ A military ATC facility may be assigned as the controlling agency if the <u>service area office</u> and the <u>concerned Air Route Traffic Control Centers (ARTCC)</u> are all in <u>agreement</u> ■ A <u>controlling agency must be designated</u> for each joint-use SUA area	The using agency is the military unit or other organization whose activity <u>established the requirement for the SUA</u>. ■ Ensures that the <u>airspace is used only for its designated purpose</u> ■ Ensures that <u>scheduling procedures</u> are established and utilized ■ Ensures the controlling agency is kept <u>informed of schedule changes and completion of activities for the day</u> ■ Ensures a <u>point of contact is made available</u> to communicate with the controlling agency about schedules, access in emergencies, weather diversions, etc.

Types of Special Use Airspace

There are 7 types of SUA:

TYPE	DESCRIPTION	EXAMPLE(S)
PROHIBITED AREA (P) 	A prohibited area is airspace established under Title 14 CFR part 73 provisions, within which no person may operate an aircraft without permission of the using agency. Prohibited areas are established when necessary to prohibit flight over an area on the surface in the interest of national security and welfare. - Established in the interest of national security and welfare - Begins at the surface of the Earth and extends upward to a specified altitude - Is depicted on aeronautical charts, identified by the letter "P," a dash, and a number	- Washington DC, U.S. Capitol and White House - Amarillo, TX, Pantex nuclear assembly plant - Naval Base Kitsap, Washington - Mount Vernon, VA, home of George Washington
RESTRICTED AREA (R) 	A restricted area is airspace established under Title 14 CFR part 73 provisions, within which the flight of aircraft, while not wholly prohibited, is subject to restriction. - Established when activities are considered hazardous to nonparticipating aircraft - Identified by the letter "R," a dash, and number on aeronautical charts	Invisible hazards to aircraft such as: - Artillery firing - Aerial gunnery - Missile activities
WARNING AREA (W) 	A warning area is airspace of defined dimensions (extending from 3NM outward from the coast of the United States), designated to contain activity that may be hazardous to nonparticipating aircraft. - Purpose is to warn nonparticipating pilots of the potential danger from activities being conducted - Identified with the letter "W" prefix followed by a dash; a two- or three-digit number - It can be located over domestic waters, international waters, or both	Missile test sites such as: Area surrounding Kauai

TYPE	DESCRIPTION	EXAMPLE(S)
ALERT AREA (A) 	An alert area is airspace wherein a high volume of pilot training or an unusual type of aeronautical activity is conducted. - Designed to inform pilots of activities they might otherwise not expect to encounter - Identified by the letter “A” prefix followed by a dash, a two- or three-digit number - Must not extend into Class A, B, C, and D airspace, or Class E airport surface areas - Should avoid Federal airways, major terminal areas, and high volume VFR routes - Only those activities that do not pose a hazard to other aircraft may be conducted in an Alert area - Must conduct activities in accordance with VFR, and in compliance with applicable Sections of Title 14 CFR	Activities such as: - Pilot training - Parachute jumping - Glider towing - Or an unusual type of aerial activity
CONTROLLED FIRING AREA (CFA) 	A controlled firing area is airspace designated to contain activities that, if not conducted in a controlled environment, would be hazardous to nonparticipating aircraft. - Provide a means to accommodate, without impact on aviation, certain hazardous activities that can be immediately suspended if a nonparticipating aircraft approaches the area - Not depicted on an aeronautical chart - Control activities that can be immediately suspended when a nonparticipating aircraft is approaching	Activities such as: - Ordnance disposal - Blasting - Static testing of large rocket motors

TYPE	DESCRIPTION	EXAMPLE(S)
MILITARY OPERATIONS AREA (MOA)  EVERS MOA	A military operations area (MOA) is airspace designated outside of Class A airspace, to separate or segregate certain nonhazardous military activities from IFR traffic and to identify for VFR traffic where these activities are conducted. - Designated to contain nonhazardous, military flight activities - Identified by a name followed by the acronym MOA	Activities such as: - Air combat maneuvers - Air intercepts - Low altitude tactics
NATIONAL SECURITY AREA (NSA) 	A national security area consists of airspace of defined vertical and lateral dimensions established at locations where there is a requirement for increased security of ground facilities. - Designated to enhance national security and protect national assets - Identified by a red dotted line surrounding the area - Have minimum dimensions that promote the protection of the national asset or area - Requested to voluntarily avoid flying through an NSA	- Areas that protect national assets - Areas that need protection in the interest of national security

Note: Pilots are requested to voluntarily avoid flying through an NSA. When it is necessary to provide a greater level of security, flight in an NSA may be temporarily prohibited pursuant to the provisions of Title 14 CFR 99.7, Special Security Instructions. Where there is a need to restrict flight operations in an NSA, the required restriction will be issued by Airspace Regulations and ATC Procedures Group and disseminated via NOTAM.



Knowledge Check C

REVIEW what you have learned so far about SUA. ANSWER the questions listed below.

1. The SUA that is necessary to confine activities considered hazardous to nonparticipating aircraft is called a _____ Area. (Select the correct answer.)
 - ☐ Prohibited
 - ☒ Restricted
 - ☐ Terminal Control
2. What type of SUA is found over international waters? (Select the correct answer.)
 - ☐ Prohibited Area
 - ☐ Controlled Firing Area
 - ☒ Warning Area
3. Prohibited Area vertical airspace begins at _____. (Select the correct answer.)
 - ☐ 3,000 feet
 - ☒ The Earth's surface
 - ☐ 1,500 feet
4. Airspace in which a high volume of pilot training takes place would be designated as a(n) _____. (Select the correct answer.)
 - ☐ Restricted Area
 - ☐ Controlled Firing Area
 - ☒ Alert Area

Special Use Airspace Summary

This lesson explained how ATC classifies and organizes Special Use Airspace and has shown how to identify Special Use Airspace, as designated in the United States.

EXERCISE: AIRSPACES

Purpose

This exercise promotes interaction and provides a review of Classes of Airspace and Special Use Airspace.

Directions

Read each question. In the column labeled Answer, put your answer.

QUESTION	ANSWER
1. Federal airways "Victor" are in which class of airspace?	E
2. Uncontrolled airspace is designated as which class of airspace?	G
3. Jet routes are in which class of airspace?	A
4. What determines the upper limits of Class G airspace?	The bottom of the next airspace
5. What is one of the requirements for an aircraft operating in Class B airspace?	ATC clearance
6. VFR flight is not permitted in which class of airspace?	A
7. Generally, that airspace from the surface to 2,500 feet above the airport elevation with an operational control tower is classified as what class airspace?	D
8. What type of SUA is over the White House?	Prohibited
9. In what type of airspace would a high volume of pilot training take place?	Alert
10. What type of SUA is found over international waters?	Warning
11. What type of SUA is over an aerial gunnery range over land?	Restricted
12. What does "MOA" stand for?	Military Operations Area

SUMMARY

The purpose of this module was to explain how Air Traffic Control (ATC) classifies, organizes, and uses classes of airspace and Special Use Airspace (SUA).

In accordance with Title 14 Code of Federal Regulations, Part 91, General Operating and Flight Rules; FAA Orders JO 7110.65, Air Traffic Control, and JO 7400.2, Procedures for Handling Airspace Matters; and the Aeronautical Information Manual (AIM), you should now be able to:

- Identify classes of airspace and their uses
- Identify SUA, as designated in the United States

KNOWLEDGE CHECK ANSWERS

Knowledge Check A

1. Class C; 2. Class B; 3. Class G; 4. Class D; 5. Class E

Knowledge Check B

1. To restrict certain aircraft from operating within a defined area on a temporary basis

Knowledge Check C

1. Restricted; 2. Warning Area; 3. The Earth's surface; 4. Alert Area